

Task 1

Find all the ports that are open on the website : <http://testasp.vulnweb.com/>

Attack Name: Port Scanning

Severity: Medium (CVSS Score: 5.3)

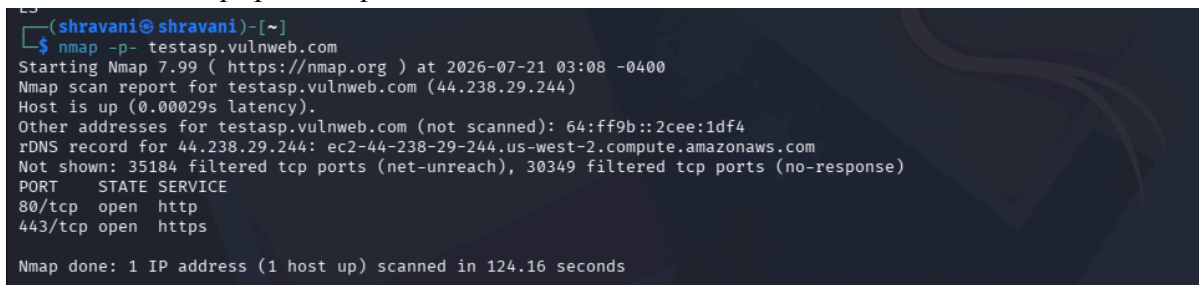
Reason:

Port scanning is an information-gathering technique used during the reconnaissance phase of a security assessment. Identifying open ports helps determine which services are running on the target system and highlights potential entry points that may require further security evaluation.

Steps to Reproduce:

Step 1: Scan the Target Using Nmap

Command: `nmap -p- testasp.vulnweb.com`



```
(shravani@shravani)-[~]
$ nmap -p- testasp.vulnweb.com
Starting Nmap 7.99 ( https://nmap.org ) at 2026-07-21 03:08 -0400
Nmap scan report for testasp.vulnweb.com (44.238.29.244)
Host is up (0.00029s latency).
Other addresses for testasp.vulnweb.com (not scanned): 64:ff9b::2cee:1df4
rDNS record for 44.238.29.244: ec2-44-238-29-244.us-west-2.compute.amazonaws.com
Not shown: 35184 filtered tcp ports (net-unreach), 30349 filtered tcp ports (no-response)
PORT      STATE SERVICE
80/tcp    open  http
443/tcp    open  https

Nmap done: 1 IP address (1 host up) scanned in 124.16 seconds
```

Fig 1: Nmap Port Scanning Results

Analysis

The Nmap scan was performed to identify the open ports and network services running on the target website. The scan revealed that the server has port 80 (HTTP) and port 443 (HTTPS) open, indicating that it supports both unsecured and secured web communication. Most of the remaining TCP ports were filtered, suggesting that they are protected by a firewall or packet filtering mechanism, reducing the exposed attack surface.

Impact

Advantages

1. Identifies the active services running on the target system.
2. Helps security professionals understand the network exposure of the server.
3. Assists in detecting unnecessary or misconfigured services that may require further assessment.

Disadvantages

1. Open web service ports can become potential attack vectors if the hosted applications contain vulnerabilities.
2. If HTTP traffic is not redirected to HTTPS, communication may be susceptible to interception.
3. Information gathered through port scanning can be used by attackers during the reconnaissance phase to plan further attacks.

Mitigation Steps

1. Redirect all HTTP traffic (Port 80) to HTTPS (Port 443) to ensure encrypted communication.
2. Keep the web server and hosted applications updated with the latest security patches.
3. Close or disable any unnecessary network services and ports that are not required.
4. Configure firewalls and access control rules to restrict unauthorized access to exposed services.
5. Perform regular vulnerability assessments and port scans to identify changes in the server's security posture.
6. Continuously monitor server logs and network activity to detect suspicious access attempts.

Resources Used: Kali Linux, Nmap and the test website <http://testasp.vulnweb.com/>.